microscopes began to reveal deeper differences that could not be seen with the naked eye. By the 1960s, picking up on an idea first proposed by Edouard Chatton in the 1930s, the need for a new division of living things emerged, placed between prokaryotes (such as bacteria, with simple nucleus-free cells), and eukaryotes (such as animals and plants with larger, more complex cells).

In the 1970s, the American biologist Carl Woese claimed that even this system failed to account for the diversity among microbes—the smallest living things. He focused on ribosomes—minuscule grains that all cells need in order to make protein—and devised what he called the "three-domain system." This gave him a new perspective on the branches of Charles Darwin's evolutionary "tree of life." Woese found big differences in the chemical makeup of ribosomes among tiny microbes, with one group as far from other prokaryotes as bacteria are from humans.

Revising the tree of life

Woese's third domain of organisms, known as archaea, is superficially similar to bacteria, but has some strange properties. Many thrive in extreme habitats. Some—uniquely among living things—generate methane in oxygen-deprived places, such as deep marine sediments, or inside warm digestive cavities, such as those of belching, flatulent plant-eating mammals. Other archaea inhabit lakes that are ten times saltier than seawater, or hot acidic pools fed by geothermal heat that would kill anything else.

A decade before Woese proposed his theory, Robert H. Whittaker had recognized animals, plants, and fungi as separate eukaryotic kingdoms, with all other eukaryotes placed in the protist kingdom, and bacteria constituting a fifth kingdom. Whittaker's protist kingdom covered eukaryotic organisms such as amoebas that did not fit the other categories. Some protists were closer to animals, some closer to plants, and others not close to either. They